NSSC 2026 · Data Analytics · IIT Kharagpur

# Unsupervised Anomaly Detection in Mars HiRISE Imagery

Convolutional Autoencoder · Isolation Forest · Reconstruction Heatmaps

Team **T-0637108** · Model **v4** · Private GitHub submission

**Highlighted final answers.** Latent dimension **512**. Loss **0.40·MSE + 0.35·(1−SSIM) + 0.25·∇L1**. Novelty −**IsolationForest.score_samples** on standardised latents (600 trees, `contamination='auto'` — does **not** set the count). Threshold $\tau$ **= 0.5259** (robust / MAD k=3.5) → **161 / 10 422** flags (~1.54%). Top-5 post-$\tau$: **sample_04586.jpg, sample_03225.jpg, sample_01380.jpg, sample_10203.jpg, sample_05057.jpg**.

$$d_{\text{lat}} = 512, \quad \tau = 0.5259, \quad N_{\text{flag}} = 161/10422$$

## 0. How this report maps to the problem statement

| Phase | Marks | Where answered |
|---|---|---|
| 1 Latent compression | 30 | §1 (arch / loss / t-SNE·UMAP) |
| 2 Isolation Forest + location | 20 | §2 (+ optional fusion §2.4) |
| 3 Heatmaps + geology | 25 | §3 |
| 4 Changelog journal | 25 | MODEL_CHANGELOG.md + §4 |

## 1. Phase 1 — Deep Latent Compression (30)

### 1.1 Architecture (10)

We train a **from-scratch** ConvAE (PyTorch). No pretrained weights, no VGG/ResNet, no transfer learning. Each 227×227 crop is reflection-padded to 228, compressed by four stride-2 blocks to a **14×14×256** map, channel-squeezed to 128, then projected to a fixed-length 1-D latent for Isolation Forest. The decoder uses nearest-neighbour upsampling + paired 3×3 convolutions (avoids transpose-conv checkerboard), bilinear resize to 227, and a sigmoid.

$$z \in \mathbb{R}^{512}$$

Fixed-length code passed to Phase 2.

**Latent-dimension tradeoff (required justification):**

| Dim | What we observed | Decision |
|---|---|---|
| 128 (v1) | Mushy dunes/rims; t-SNE nearly one blob | Reject — under-expressive |
| 256 (v2) | Better global structure; soft ridgelines | Better, still soft |
| **512 (v3/v4)** | Usable heatmaps; IF remains separable | **Selected** |

### 1.2 Loss design (10)

Pure MSE over-smooths high-frequency Martian texture. We use a fully differentiable mix of MSE, SSIM, and finite-difference gradient L1 — **without** a pretrained perceptual network (disallowed by the brief). Training follows a curriculum (MSE warmup → add SSIM/grad → raise weights → edge fine-tune), then a v4 continue-train at lr=5×10■■ with the final coefficients.

$$\mathcal{L} = 0.40\,\text{MSE} + 0.35\,(1 - \text{SSIM}) + 0.25\,\|\nabla x - \nabla \hat{x}\|_1$$

Final objective.

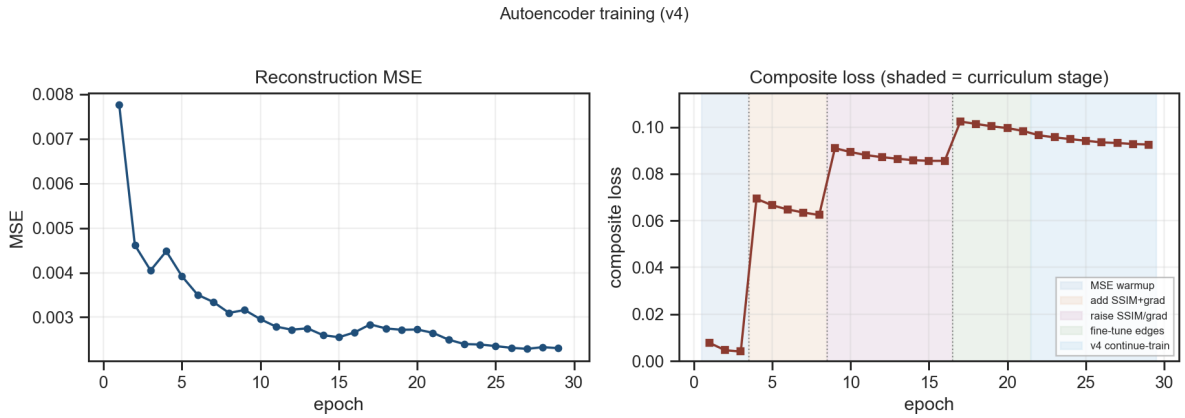



Fig. 1 — Training log. Shaded bands = curriculum stages; late band = v4 continue-train.

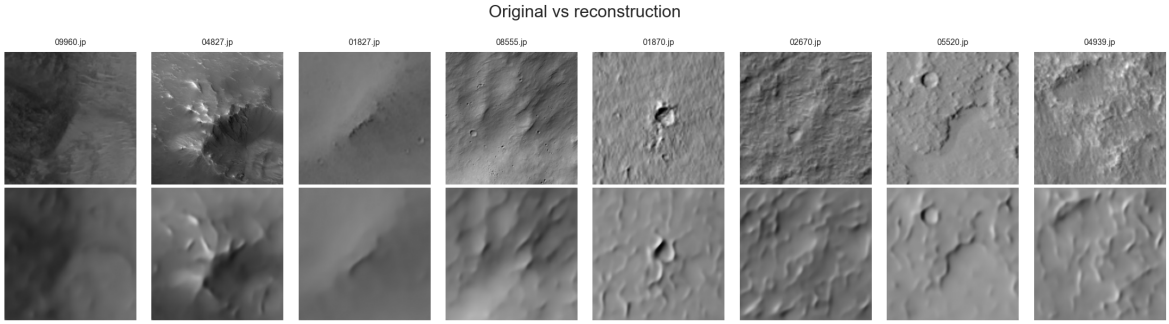



Fig. 2 — Live-style originals vs reconstructions (large-scale landforms recovered).

## 1.3 Latent visualisation (10)

t-SNE and UMAP are **qualitative diagnostics**, not geological class labels and not proof of encoder collapse. Colour = Phase-2 novelty. A dense low-novelty core with higher-novelty points on the fringe is the healthy pattern we look for before trusting Isolation Forest.

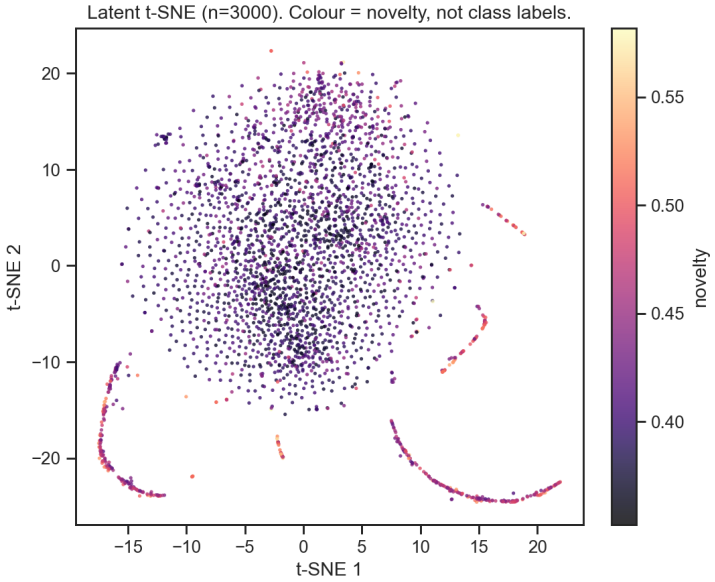



Fig. 3 — t-SNE of standardised latents.

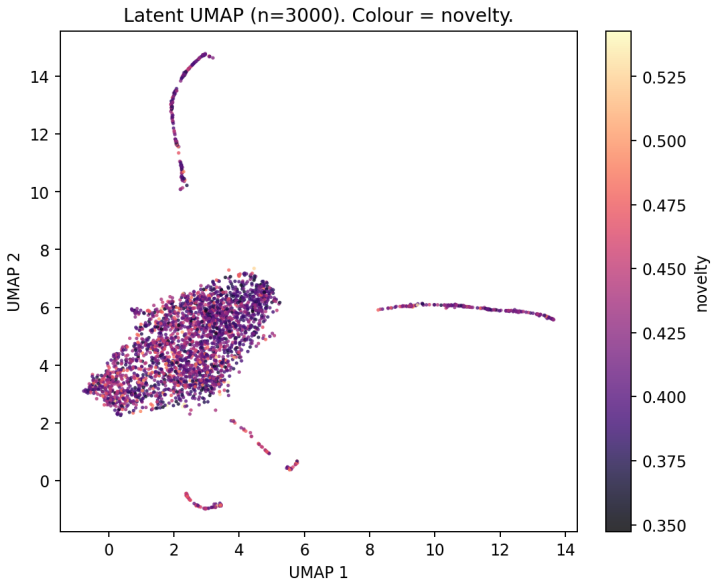



Fig. 3b — UMAP diagnostic (same subsample policy).

## 2. Phase 2 — Isolation Forest Novelty Engine (20)

### 2.1 Novelty scoring (5)

Isolation Forest (600 trees, `contamination='auto'`) is fit on **standardised 512-D latents**, not raw pixels. Sklearn's native `score_samples` is higher for inliers; we invert it to match the competition convention (higher novelty ⇒ more anomalous).

$$s = -\text{score_samples}(\tilde{z}), \quad \tilde{z} = \text{StandardScaler}(z)$$

### 2.2 Statistical thresholding (9)

**Rejected:** fixed top-N cutoffs. **Compared on the observed score distribution:** (i) elbow on the sorted novelty curve (rank 520, thr≈0.4930); (ii) Gaussian reference μ+2.5σ≈0.5116 (histogram is skewed — Gaussian used only as a check); (iii) robust MAD / modified-z with k=3.5. **Selected: τ=0.5259 (robust)** → **161** flags. Contamination never determines this count.

$$\tau_G = \mu_s + 2.5\,\sigma_s$$

$$\tau_{MAD} = \text{median}(s) + \frac{3.5}{0.6745}\,\text{MAD}(s)$$

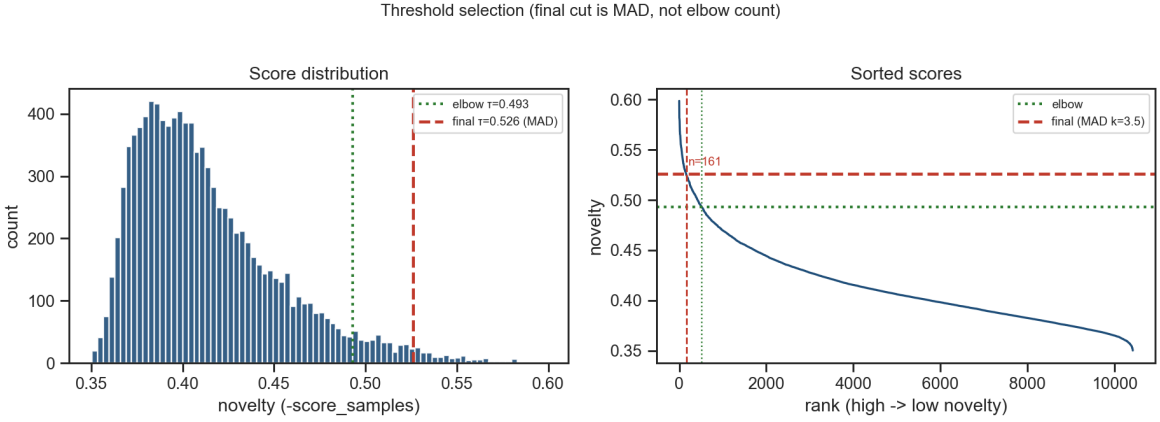



Fig. 4 — Histogram + sorted novelty with elbow vs final MAD cut.

### 2.3 Image location analysis (6)

Each crop is joined to `source_image_metadata.csv` via `source_image_id` from `crop_metadata_index.csv`. In the released table, longitude only takes values **0° and 180°**, so a lon–lat scatter is not a planetary map. We therefore compare flagged vs all on latitude, sun angle, and season, and keep a jittered coordinate panel for transparency.

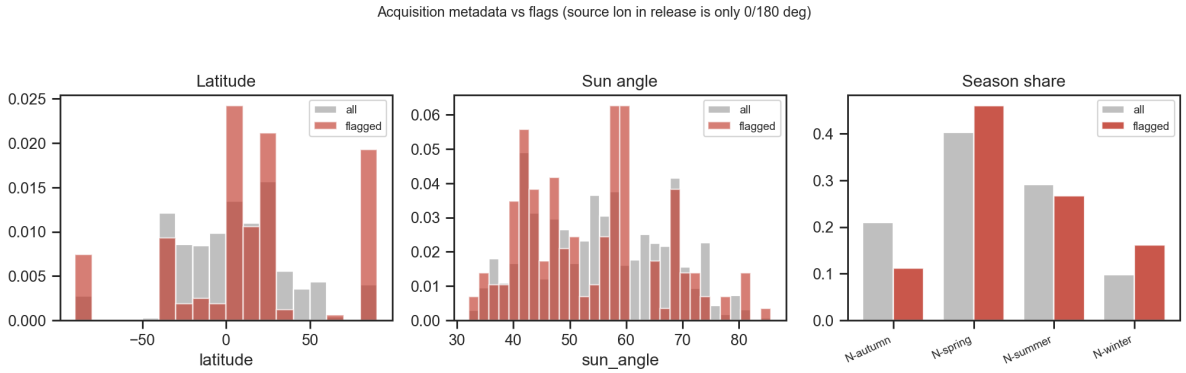



Fig. 5 — Acquisition covariates for flagged vs full set.

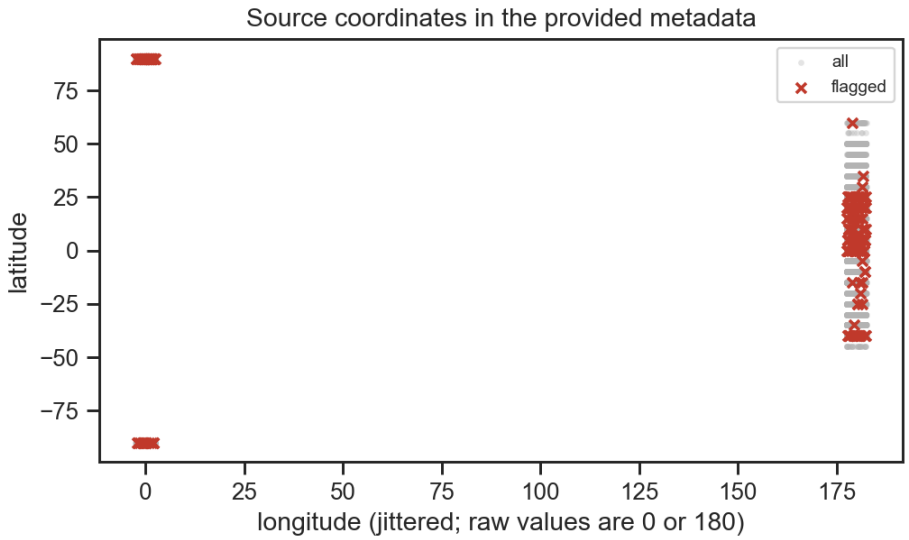



Fig. 5b — Source coordinates (longitude jittered).

## 2.4 Optional metadata fusion (+2) & recon hybrid

Metadata fusion concatenates the image latent with normalised sun_angle / lat / lon / resolution / season one-hots and re-fits Isolation Forest. Fused μ+2.5σ flags **266** (overlap with image-only: **160**). A separate recon hybrid `0.75·z(IF)+0.25·z(MSE)` flags **65** (overlap **53**). **Primary Phase-2 answer remains image-latent IF + MAD.**

$$u = [\, z;\ \tilde{\theta}_{\text{sun}};\ \tilde{\phi}_{\text{lat}};\ \tilde{\lambda}_{\text{lon}};\ \tilde{r};\ \mathbf{e}_{\text{season}} \,]$$

$$s_{\text{hyb}} = 0.75\, z(s_{\text{IF}}) + 0.25\, z(\text{MSE}_{\text{recon}})$$

| Pipeline | Cutoff | Flags |
|---|---|---|
| Primary IF + MAD | τ=0.5259 | 161 |
| Metadata fused | μ+2.5σ=0.5153 | 266 |
| Recon hybrid | MAD/elbow on s_hyb | 65 |

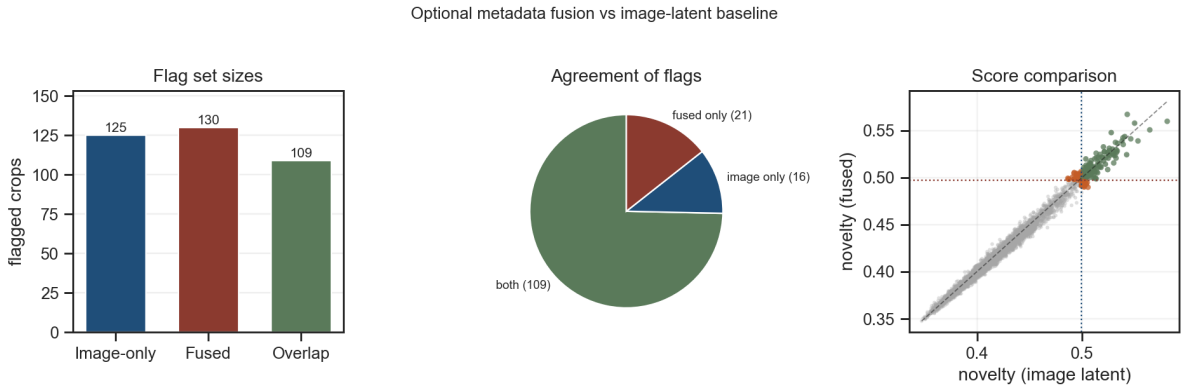



Fig. 7 — Metadata fusion vs image-only comparison.

# 3. Phase 3 — Reconstruction Interpretability (25)

## 3.1 Pixel-wise error heatmaps (15)

After applying $\tau$=0.5259, we rank crops with novelty $\geq \tau$ and select the **top five**. “Top-5” is a selection step **after** statistical thresholding — never a substitute for it. Heatmaps show per-pixel absolute error through the Phase-1 decoder (left column = original).

$$E_{ij} = \left| x_{ij} - \hat{x}_{ij} \right|$$

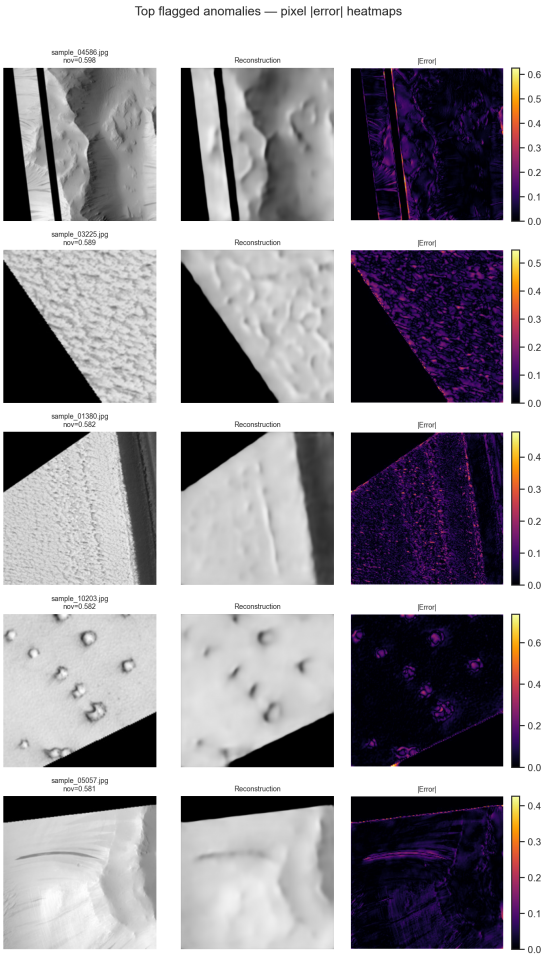



Fig. 6 — Top-5 post-threshold: input · reconstruction · |error|.

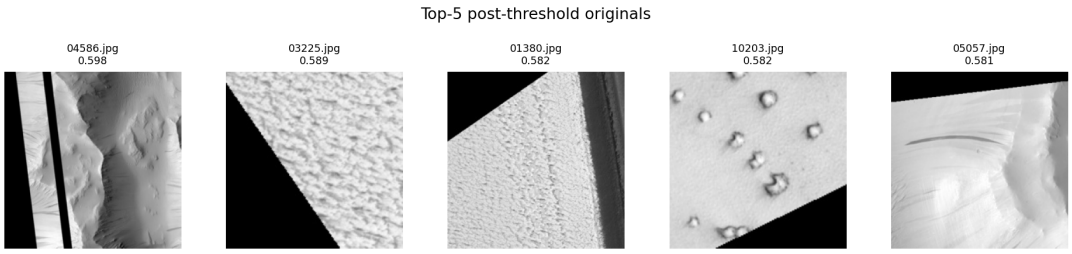



Fig. 6b — Same five originals at a glance.

### 3.2 Geological report (10) — hypotheses only

`sample_04586.jpg` (novelty=0.598, SRC_022): Hard black vertical strip; residual energy along the strip edge → **hypothesis: crop/sensor framing mask or readout bar**, not a natural landform.

`sample_03225.jpg` (novelty=0.589, SRC_039): Granular field + dark corner; **diffuse frame-wide** residual → **hypothesis: domain-shifted / atypical texture** (terrestrial-like content or strong corruption).

`sample_01380.jpg` (novelty=0.582, SRC_039): Related source with concentrated structural residual → **hypothesis: shared artifact mode or spliced patch** on SRC_039 material.

`sample_10203.jpg` (novelty=0.582, SRC_166): Compact circular mounds; error **localises on those blobs** → **hypothesis: semantic splice** of incompatible patches, or rare under-represented mounds.

`sample_05057.jpg` (novelty=0.581, SRC_146): Thin horizontal slash erased in reconstruction and lit in the residual → **hypothesis: sensor streak / glitch line**.

## 4. Phase 4 — Design journal (25)

The primary Phase-4 artifact is `MODEL_CHANGELOG.md`, with **symptom** → **diagnosis** → **fix** entries for **v1** → **v2** → **v3** → **v4**. Briefly: v1 (MSE, 128) over-smoothed; v2 (deeper+SSIM, 256) improved structure but left thresholding weak; v3 kept a 14×14 map, latent 512, MSE+SSIM+∇L1, MAD cutoff + metadata fusion; v4 restored the edge fine-tune path, continue-trained, and added a comparative recon hybrid. Iterations need not monotonically “win”; each change is motivated by an observed failure mode.

## 5. Summary of answers (highlighted)

| Item | Answer |
|---|---|
| Model version | **v4** |
| Latent dimension | **512** |
| Loss (final) | **0.40 / 0.35 / 0.25** (MSE, 1–SSIM, ∇L1) |
| Novelty definition | –**IF.score_samples** (↑ anomalous) |
| IF contamination | `auto` — count from MAD τ |
| Threshold τ | **0.525887** (robust, k=3.5) |
| Flagged count | **161 / 10422** |
| Top-5 (post-τ) | sample_04586.jpg, sample_03225.jpg, sample_01380.jpg, sample_10203.jpg, sample_05057.jpg |

Accompanying repository material: `HiRISE_Anomaly_Detection.ipynb` (PS-numbered cells with live computation), `MODEL_CHANGELOG.md`, `submission/` (checkpoint, figures, scored CSV, threshold JSON).